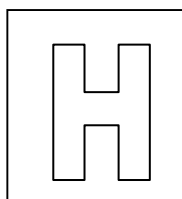


Candidate Name: _____

Class Adm No

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2018 Preliminary Exams Pre-University 3

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

17 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer all questions in the space provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Section B	
Total	

This question paper consists of 21 printed pages, including 1 blank page.

[Turn over

Section A

Answer **all** questions in this section.

1. Monoclonal antibodies are antibodies derived from a single unique parent cell. In the industry, monoclonal antibodies are produced from hybridoma. The procedures are illustrated in Fig. 1.1.

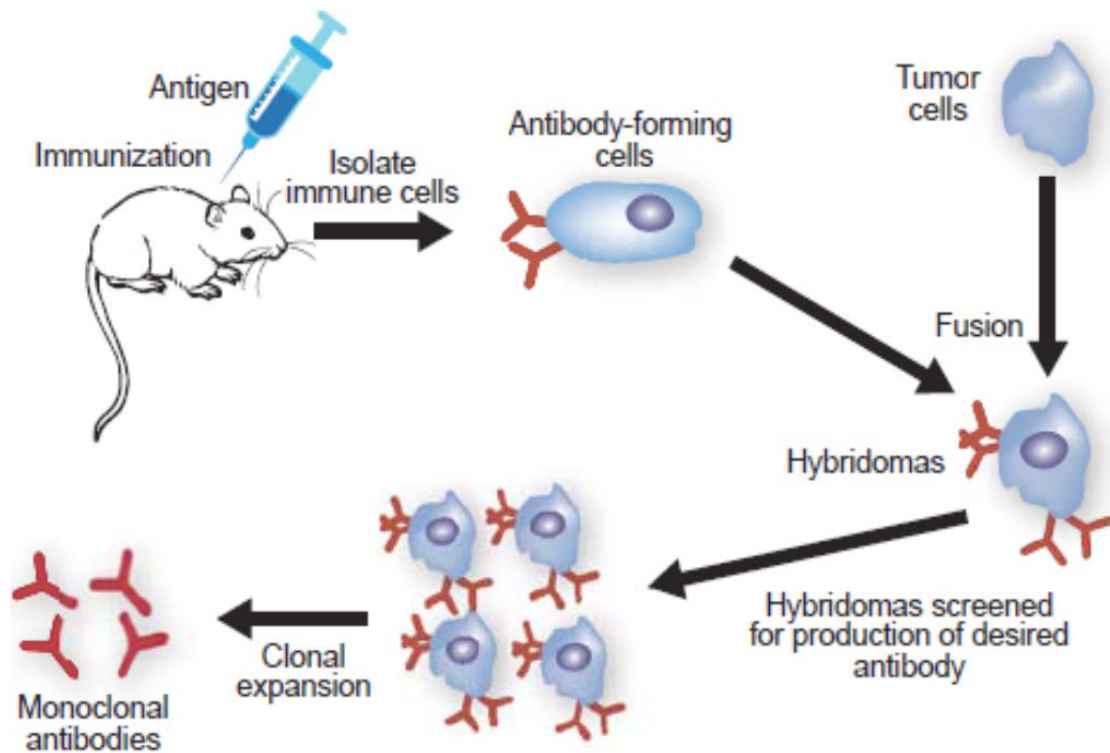


Fig. 1.1

The steps to produce monoclonal antibodies using hybridoma technique are as follow:

- Step 1: A mouse is injected with a specific antigen.
- Step 2: The mouse is left for a few weeks to allow immune response to occur.
- Step 3: Immune cells are isolated and plasma cells are extracted from the mouse's spleen.
- Step 4: Plasma cells are fused with tumour cells to form hybridoma.
- Step 5: Hybridomas are screened to select the ones that are producing the desired antibody.
- Step 6: Selected hybridomas are isolated to grow and divide during clonal expansion.

- (a) State the type of immunity involved in the production of monoclonal antibody.

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- (b) In the space below, draw a labelled diagram of an antibody such as immunoglobulin G (IgG).

[3]

- (c) Describe the events that occur in step 2 during the immune response in the hybridoma technique.

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Hybridoma is a hybrid cell consisting of both tumour and plasma cells.

(d) Explain how mutations result in cancer.

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Besides genetic mutation, cancer can be caused by other causative factors.

(e) Identify two such factors.

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The tumour cells were genetically modified such that an enzyme known as hypoxanthine-guanine-phosphoribosyltransferase (HGPRT) is absent. HGPRT enzyme is involved in the synthesis of purine. On the other hand, the plasma cells have the HGPRT enzyme. Therefore, the fused plasma cell in the hybridoma supplies the HGPRT enzyme for the survival of the hybridomas.

- (f)** Explain the roles of purine in living organisms.

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- (g)** Explain why unfused tumour cells and unfused plasma cells are unable to grow for a long period of time.

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The HGPRT enzyme consists of four polypeptide chains, as shown in Fig. 1.2.

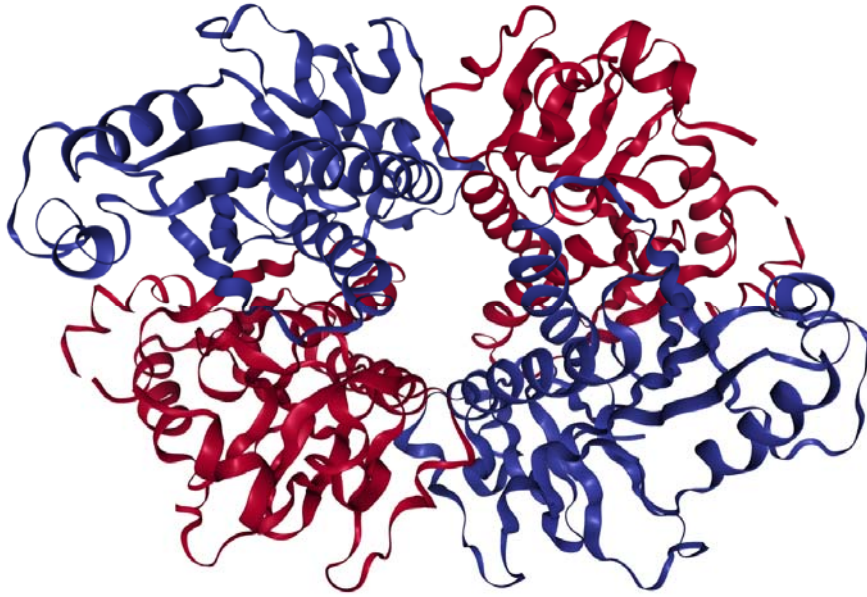


Fig. 1.2

(h) Explain the importance of the shape of HGPRT enzyme to its function.

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Rheumatoid arthritis (RA) is an autoimmune disease in which T lymphocytes attack the cartilage of joints by secreting a protein, $\text{TNF}\alpha$. RA can be treated using either the anti-inflammatory drug, MTX, or monoclonal antibody, infliximab.

Five groups of people with RA received the following treatments for one year:

- group **P** – MTX only
- group **Q** – MTX plus low dosage of infliximab at intervals of eight weeks
- group **R** – MTX plus low dosage of infliximab at intervals of four weeks
- group **S** – MTX plus high dosage of infliximab at intervals of eight weeks
- group **T** – MTX plus high dosage of infliximab at intervals of four weeks.

At the end of the year's treatment, the proportion of people in each group with increased joint damage was determined.

The experimental results are shown in Fig. 1.3. The number of people in each group is shown in brackets. All bar graphs display means $\pm$ standard error mean (S.E.M) (sample size ≥ 3). Comparisons between infliximab treatments and untreated controls were made using a Student's t-test (* $p < 0.5$, ** $p < 0.01$, *** $p < 0.001$).

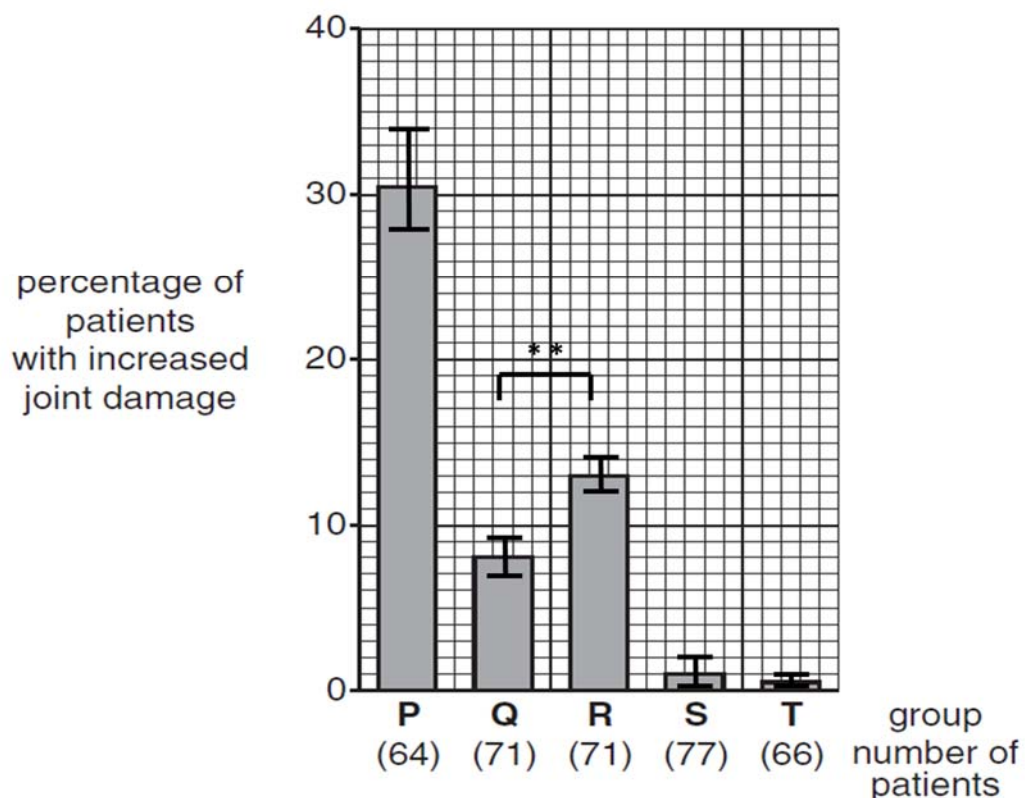


Fig. 1.3

- (i) Discuss the effectiveness of infliximab treatment on these patients.

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- (j) With reference to the S.E.M and the t-test results, assess the validity of the experimental results.

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[Total:25]

2. The middle of the Nevada desert seems an unlikely spot to find a fish. But an underground fissure in scorching-hot Death Valley is the only natural habitat for the endangered Devils Hole pupfish, a silvery blue creature about the size of a pet goldfish.

The water inside Devils Hole is consistently a toasty 32°C, hot enough to kill most other fish. For two months of the year, the cavern—which opens into a deep subterranean lake—receives no direct sunlight. And yet scientists believe a small population of pupfish has lived there for 10,000 to 20,000 years.

Now, evidence is growing that these fish might be far younger than previously assumed: A new study suggests that the Devils Hole pupfish actually colonised its watery cavern somewhere between 105 and 830 years ago, making scientists rethink how it got there in the first place. “Our new estimate of the younger age in a way makes this species even more fascinating because it has so many of these unique traits relative to other pupfish,” says Christopher Martin, a biologist at the University of North Carolina, Chapel Hill, who led the study.

Pupfish first arrived in Death Valley during a wet period, probably when water joined the area to lakes or rivers elsewhere. But as the water dried up and the area turned to desert, the pupfish became isolated in a smattering of springs.

Devils Hole is especially isolated, and the pupfish population there is particularly small, ranging from 35 to 548 fish in the remote cavern since official recordkeeping began in the 1970s. Their survival has been heralded as an evolutionary anomaly, as such populations usually become severely inbred and die out. “All we know about conservation genetics suggests that these populations of very small size should not be able to survive in the long term,” Martin says.

To figure out just when the Devils Hole pupfish diverged from its kin, the researchers sequenced 13,000 different stretches of DNA from 56 pupfish from around Death Valley and the world. Those data allowed them to reconstruct the area's pupfish family tree and calculate when the different species emerged. Although scientists previously believed the first pupfish species came to Death Valley several million years ago, these analyses suggest they arrived around the time of the valley's most recent flood, just 10,000 years ago. The analysis also suggests the Devils Hole pupfish became isolated from other pupfish in Death Valley fewer than a thousand years ago, much more recently than expected.

<http://www.sciencemag.org/news/2016/01/bizarre-desert-dwelling-fish-may-have-evolved-just-couple-hundred-years-ago>. Extracted on 10th June 2018.

- (a)** Suggest why there is discrepancy to the age of the early ancestor of pupfish.

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Scientists believed that the change from an extensive lake system to a few pools drives the evolution of five species of the pupfish.

- (b)** Explain how this may happen.

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During autumn in 2017, there were only 115 of the fishes in the Death Valley.

- (c)** Suggest why the pupfish becomes endangered when their population falls to very low numbers.

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In the study, the scientists used biochemical data to establish the evolutionary relationship of the pupfish.

- (d)** Describe two other methods that the scientists could use to understand the homology between different fish species.

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The pupfish survives in water with high temperature of 32°C, minimal food, and receive no sunlight two months of the year.

- (e)** Describe the effect of temperature on protein structure and function.

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- (f)** Explain how the biomolecules of organisms adapt to the harsh living condition of Death Valley.

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The Death Valley receives less than two inches of rain every year. In comparison, the average rainfall in the United States is 32 inches in 2017.

- (g)** Predict what may happen to the pupfish if water level rises to form an extensive lake system.

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Rising sea level may be caused by the addition of extra water from ice sheet or the thermal expansion of sea water. Thermal expansion of sea water refers to the increase in volume of sea water due to the increased sea temperature.

- (h)** Describe three other effects of increased sea temperature.

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Besides the increase in sea temperature, another effect of global warming is the increase in surface temperature. Fig. 2.1 shows the annual mean surface temperature in Singapore from 1948 to 2011.



Fig. 2.1

- (i) Describe the changes in annual mean surface temperature from 1948 to 2011.

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- (j) Suggest the event that may take place in 1975.

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[Total:25]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 3 (a)** With reference to a named virus and its reproductive cycle, describe how the virus can transfer genes between bacteria. **[10]**

- (b)** 'Without genetic variations, antibody is completely useless in the immune system.'

Discuss the validity of this statement. **[15]**

[Total: 25]

- 4 (a)** Plants and other photosynthetic organisms convert light energy into chemical energy in the form of ATP and NADPH. Photosynthesis comprises both light and dark reactions, which are catalysed and regulated by proteins.

Describe the importance of proteins in photosynthesis. **[10]**

- (b)** Discuss the extent to which increased temperature can lead to the increase in dengue transmission. **[15]**

[Total: 25]

End of Paper